

Structural Estimation in Practice

Identification, Moments, Likelihood, Simulation, and Fit
Empirical Methods — Week 7

Teaching deck draft

Week 7 question

- How do researchers take a structural model to data credibly?
- Which variation identifies the latent primitives?
- Which likelihood pieces, moments, or auxiliary statistics carry the identifying power?
- How should fit, counterfactual discipline, and uncertainty be reported?

Structural credibility is the chain: identification $\rightarrow$ estimation $\rightarrow$ fit $\rightarrow$ counterfactual discipline $\rightarrow$ inference.

Identification in structural models

Question	Applied object	Credibility check
What is latent?	preferences, costs, beliefs, technology, risk	does the paper name the primitive?
What is observed?	choices, states, outcomes, prices, rules	are the data rich enough?
What moves it?	policy variation, prices, histories, transitions	which variation shifts the primitive?
What else could explain it?	omitted state, heterogeneity, expectations	what breaks observational equivalence?
What uses it?	welfare or policy counterfactual	is the counterfactual near support?

$$\hat{h}_N(Y, X, S, P) \iff h(\theta; \mathcal{M})$$

Likelihood vs moments vs simulation

Likelihood

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta} \sum_i \log f_{\theta}(y_i \mid x_i, s_i, p_i)$$

GMM / method of moments

$$\hat{\theta}_{\text{GMM}} = \arg \min_{\theta} \hat{g}_N(\theta)' W_N \hat{g}_N(\theta), \quad \hat{g}_N(\theta) = \hat{m}^{data} - m^{model}(\theta)$$

SMM

$$\hat{\theta}_{\text{SMM}} = \arg \min_{\theta} [\hat{m}^{data} - \hat{m}^{sim}(\theta; R)]' W_N [\hat{m}^{data} - \hat{m}^{sim}(\theta; R)]$$

$$\hat{\theta}_{\text{II}} = \arg \min_{\theta} \left[\hat{a}^{data} - \hat{a}^{sim}(\theta) \right]' W_N \left[\hat{a}^{data} - \hat{a}^{sim}(\theta) \right]$$

- The structural model generates simulated histories.
- The auxiliary model summarizes behavior in a familiar empirical language.
- The auxiliary statistics must capture the behavioral margins that matter.
- Good auxiliary fit is not enough if the statistic is unrelated to the counterfactual.

Weighting matrices and targeted moments

$$Q_N(\theta) = \widehat{g}_N(\theta)' W_N \widehat{g}_N(\theta), \quad W^* = S^{-1}, \quad S = \mathbb{E}[g_i(\theta_0)g_i(\theta_0)']$$

$$J_N = N \widehat{g}_N(\widehat{\theta})' \widehat{W}_N \widehat{g}_N(\widehat{\theta})$$

- Weighting determines which discrepancies the criterion punishes.
- Targeted moments define the model's empirical obligations.
- Overidentification lets the model fail, but non-rejection is not validation.
- Report moment scaling, weights, dominated moments, and sensitivity.

Fit, validation, counterfactual discipline

Object	What to show	Why it matters
Targeted fit	empirical vs model moments	criterion transparency
Untargeted fit	distributions, transitions, hold-out moments	validation beyond construction
Comparative statics	response to prices, policies, states	economic mechanism
Counterfactual distance	support, policy range, state distribution	extrapolation risk
Sensitivity	weights, calibrated values, seeds, starts	numerical and model dependence

$$\hat{\theta} \implies \{\sigma_{\hat{\theta}}(a \mid s; p), F_{\hat{\theta}}(s' \mid s, a; p), Y_{\hat{\theta}}(p), W_{\hat{\theta}}(p)\}$$

Variance, bootstrap, delta method

Likelihood / sandwich

$$\widehat{\text{Var}}(\hat{\theta}) \approx \frac{1}{N} \hat{A}^{-1} \hat{B} \hat{A}^{-1}, \quad \hat{B} = \frac{1}{N} \sum_i s_i(\hat{\theta}) s_i(\hat{\theta})'$$

GMM

$$\text{Avar}(\hat{\theta}) = \frac{1}{N} (G'WG)^{-1} G'WSWG (G'WG)^{-1}$$

Delta method

$$\widehat{\text{Var}}(\psi(\hat{\theta})) \approx \hat{D}_\psi \widehat{\text{Var}}(\hat{\theta}) \hat{D}_\psi'$$

Bootstrap

$$\hat{\theta}^{*(b)} = \hat{\theta}(Y_1^{*(b)}, \dots, Y_N^{*(b)})$$

Dependence and simulation error

- Cluster or block resampling should follow the independent source of variation.
- Panel histories need resampling schemes that preserve state paths.
- Market or policy moments often require market-level or policy-level clustering.
- SMM and indirect inference need seed checks and enough simulation draws.
- Generated regressors and two-step objects require correction or full-pipeline bootstrap.
- Report the uncertainty for the counterfactual object, not only primitive parameters.

Theory to applied examples

Paper	Question	Estimation lesson	Fit / counterfactual lesson
Rust	replacement timing	dynamic likelihood	hazards by state; cost counterfactuals
BLP / Nevo	differentiated demand	moments, instruments, simulation	substitution before policy claims
Todd-Wolpin	schooling subsidy	dynamic model with validation	experiment disciplines simulation
Low-Meghir-Pistaferri	lifecycle risk	moments for risk and insurance	lifecycle fit and welfare
Adda-Dustmann-Stevens	career costs of children	dynamic lifecycle structure	long-run state-specific paths

Limitations and credibility threats

- Identification by assumption: key primitives move because of normalization, not data.
- Weak moments: targeted objects do not move the primitive enough.
- Weighting dependence: conclusions change when moment weights change.
- Simulation noise: the optimizer chases random draws or local minima.
- Off-target failure: the model matches means but misses distributions or transitions.
- Counterfactual extrapolation: the policy changes support, menus, beliefs, or equilibrium conditions.
- Underreported uncertainty: first-stage, clustering, or simulation error is ignored.

Reproduce

- Rust-style replacement
- likelihood objective
- GMM moments
- targeted fit table

Diagnose

- weighting sensitivity
- untargeted support
- cluster bootstrap
- delta-method
counterfactual

Transfer

- lifecycle schooling/work
- SMM moment fit
- tuition counterfactual
- validation memo

- Estimation strategy is part of the identification argument.
- Likelihood, moments, SMM, and indirect inference differ in what they require the model to explain.
- Weighting and targeting choices should be reported as substantive design choices.
- Fit is useful only when tied to the counterfactual and shown beyond construction.
- Inference must match the estimator, dependence structure, simulation design, and policy object.